

Synopsis

Smart Employee Productivity & Burnout Prediction System Using Machine Learning

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1. Introduction

Employee productivity and employee well-being are among the most significant factors influencing organizational success. Modern organizations generate vast amounts of workforce data including attendance, workload, employee performance, project assignments, leave records, working hours, and appraisal information. While this data is continuously collected, most organizations utilize it only for operational purposes rather than extracting predictive insights.

Employee burnout has become a major organizational concern. Excessive workload, prolonged working hours, insufficient work-life balance, and occupational stress often reduce productivity and increase employee turnover. Traditional HR systems generally focus on storing employee records and generating reports but provide little assistance in identifying employees who are at risk of burnout before productivity begins to decline.

The proposed **Smart Employee Productivity & Burnout Prediction System Using Machine Learning** addresses this challenge by integrating Machine Learning with a modern web-based employee management platform. The application enables organizations to manage employee information while simultaneously predicting burnout risk and providing actionable insights through dashboards and analytical reports.

2. Problem Statement

Organizations generally rely on periodic performance reviews and manual observation to evaluate employee well-being. Such approaches are subjective and often fail to identify burnout at an early stage. The absence of predictive analytics prevents Human Resource departments from taking proactive measures to

improve employee health and organizational productivity.

The proposed system aims to automate employee management while using Machine Learning techniques to predict burnout risks using historical employee information.

3. Objectives

The primary objectives of the project are:

- Develop a secure employee management application.
 - Implement role-based authentication.
 - Maintain employee and department information.
 - Record attendance and performance data.
 - Predict employee burnout using Machine Learning.
 - Generate dashboards and analytical reports.
 - Improve organizational decision making using predictive analytics.
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4. Scope

The application supports:

- Employee Management
- Department Management
- Attendance Tracking
- Performance Monitoring
- Burnout Prediction
- Reporting and Analytics
- Administrative Dashboard
- Secure Authentication

The solution is intended for organizations seeking an intelligent Human Resource Analytics platform.

5. Existing System

Traditional employee management applications primarily provide CRUD operations, attendance management, payroll support, and report generation. These systems generally lack intelligent decision-support capabilities and cannot predict employee burnout or productivity trends.

6. Proposed System

The proposed application combines an Angular-based frontend with a FastAPI backend, PostgreSQL database, and Machine Learning model to provide predictive HR analytics.

Major capabilities include:

- Secure Login
 - Employee Management
 - Department Management
 - Attendance Monitoring
 - Productivity Analysis
 - Burnout Prediction
 - Dashboard Visualization
 - RESTful API Integration
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7. Technology Stack

Layer	Technology
Frontend	Angular
Backend	FastAPI (Python)
Database	PostgreSQL
Machine Learning	Scikit-learn
Authentication	JWT
API	REST

8. Project Modules

- Authentication Module
 - Employee Module
 - Department Module
 - Attendance Module
 - Dashboard Module
 - Burnout Prediction Module
 - Reporting Module
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9. Resources Required

Hardware

- Intel Core i5 or above
- Minimum 8 GB RAM
- 20 GB Free Storage

Software

- Python
 - Angular
 - PostgreSQL
 - Visual Studio Code
 - Git
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10. Expected Outcome

The completed application provides a secure, scalable, and intelligent HR analytics platform capable of predicting employee burnout, monitoring workforce productivity, and assisting organizations in making proactive decisions.

11. Project Deliverables

- Angular Web Application
 - FastAPI Backend
 - PostgreSQL Database
 - Machine Learning Model
 - REST APIs
 - Technical Documentation
 - Project Report
 - Presentation
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12. Conclusion

The Smart Employee Productivity & Burnout Prediction System demonstrates the practical integration of Software Engineering principles, Artificial Intelligence, Machine Learning, Database Management Systems, and Full-Stack Web Development. The proposed solution improves organizational decision-making by

enabling early identification of burnout risks and providing meaningful workforce analytics.